F FWallSim · Firewall Course

CLASS 1

System basics

GUI, System settings, local-in

Beginner | ~10 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

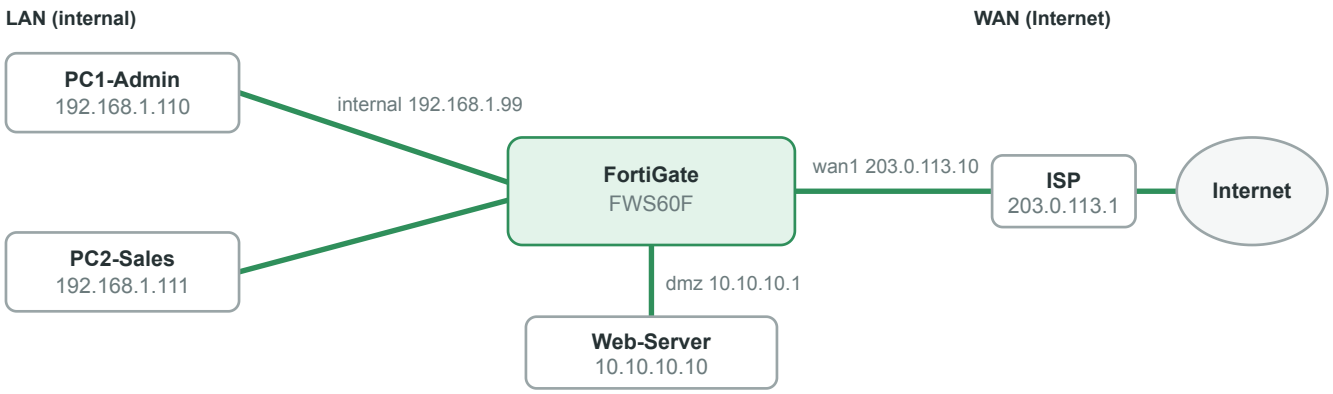
YOU WILL LEARN

- GUI
- System settings
- local-in

BEFORE YOU START

Start FWallSim with `npm run dev` and open http://localhost:5173.

The lab network



The goal

THE IDEA

Get comfortable with the GUI. You will rename the firewall, change an admin setting, and prove you can reach the FortiGate from the LAN.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **System basics** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **Change the hostname to HQ-FGT**

   System > Settings > Host name, then Apply. Watch the header and the CLI prompt change.

2. **Set the admin idle timeout to 15 minutes**

   System > Settings > Idle timeout.

3. **From PC1, ping the FortiGate (192.168.1.99)**

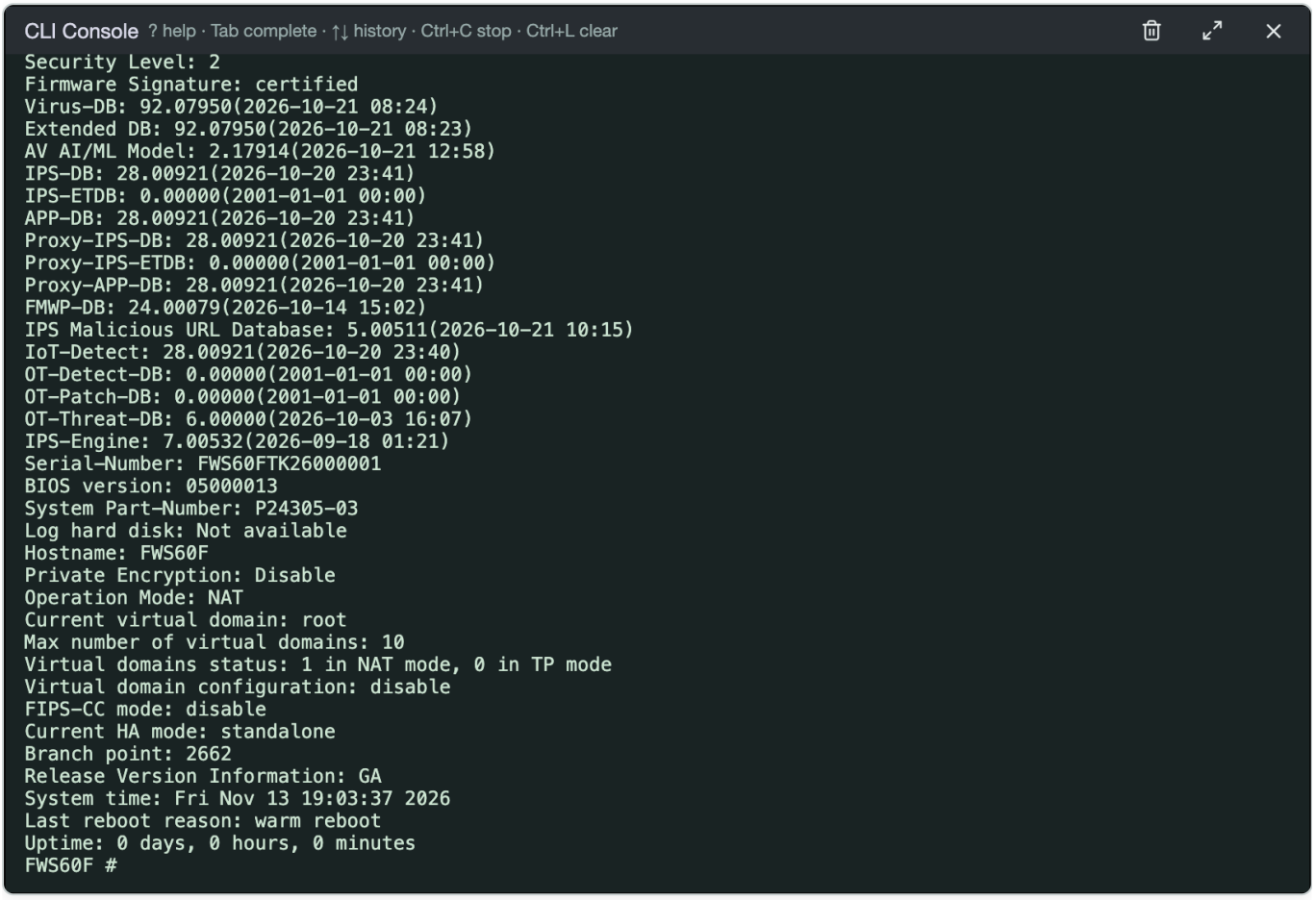
   Lab > Traffic Generator: Source PC1-Admin, Destination 192.168.1.99, application Ping, Send traffic. The Packet Walk shows it is local-in traffic.

4. **Open the CLI and run "get system status"**

   Click the >_ button in the header, type get system status and press Enter.

Screenshots

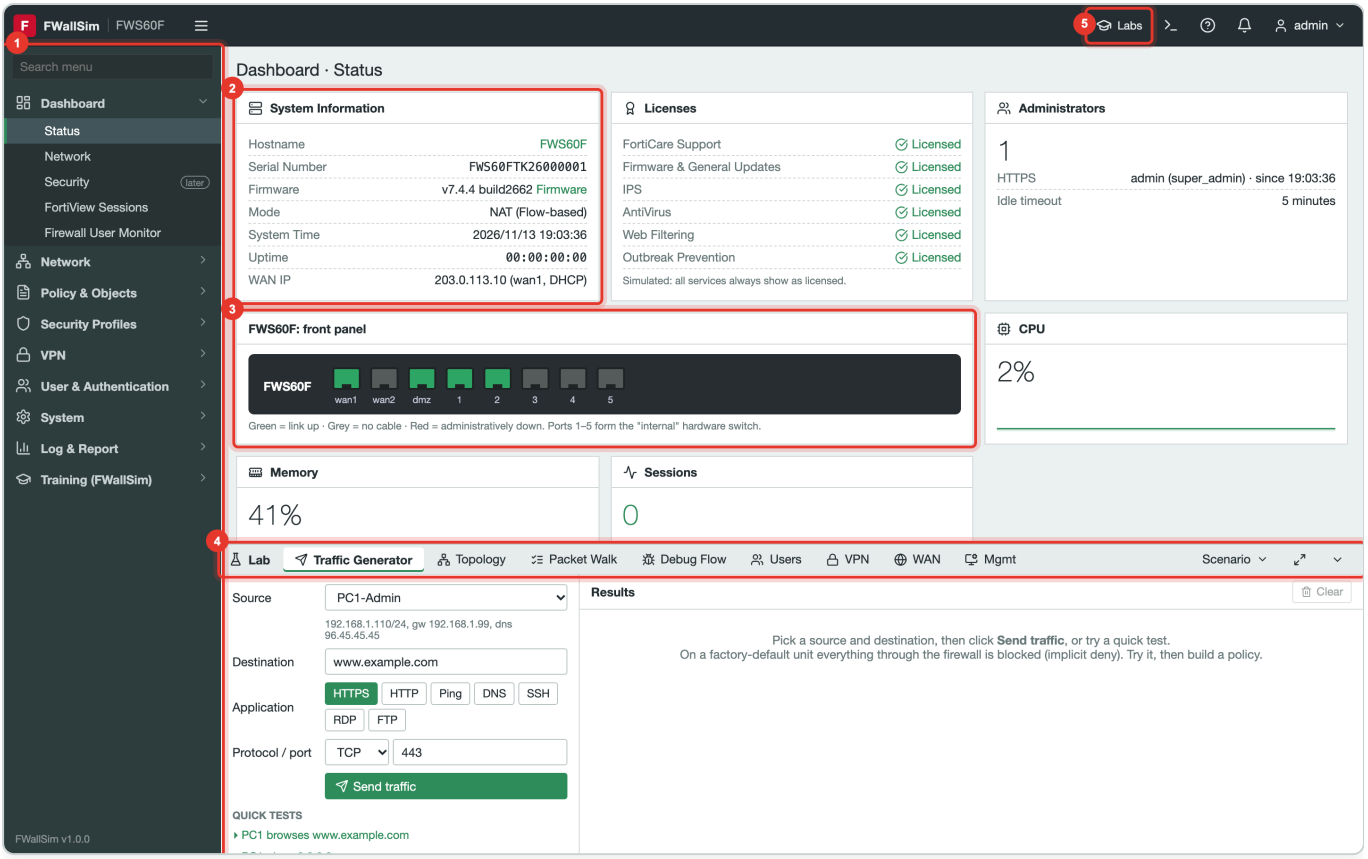
Numbered red markers show where to click or what to look at.



CLI Status



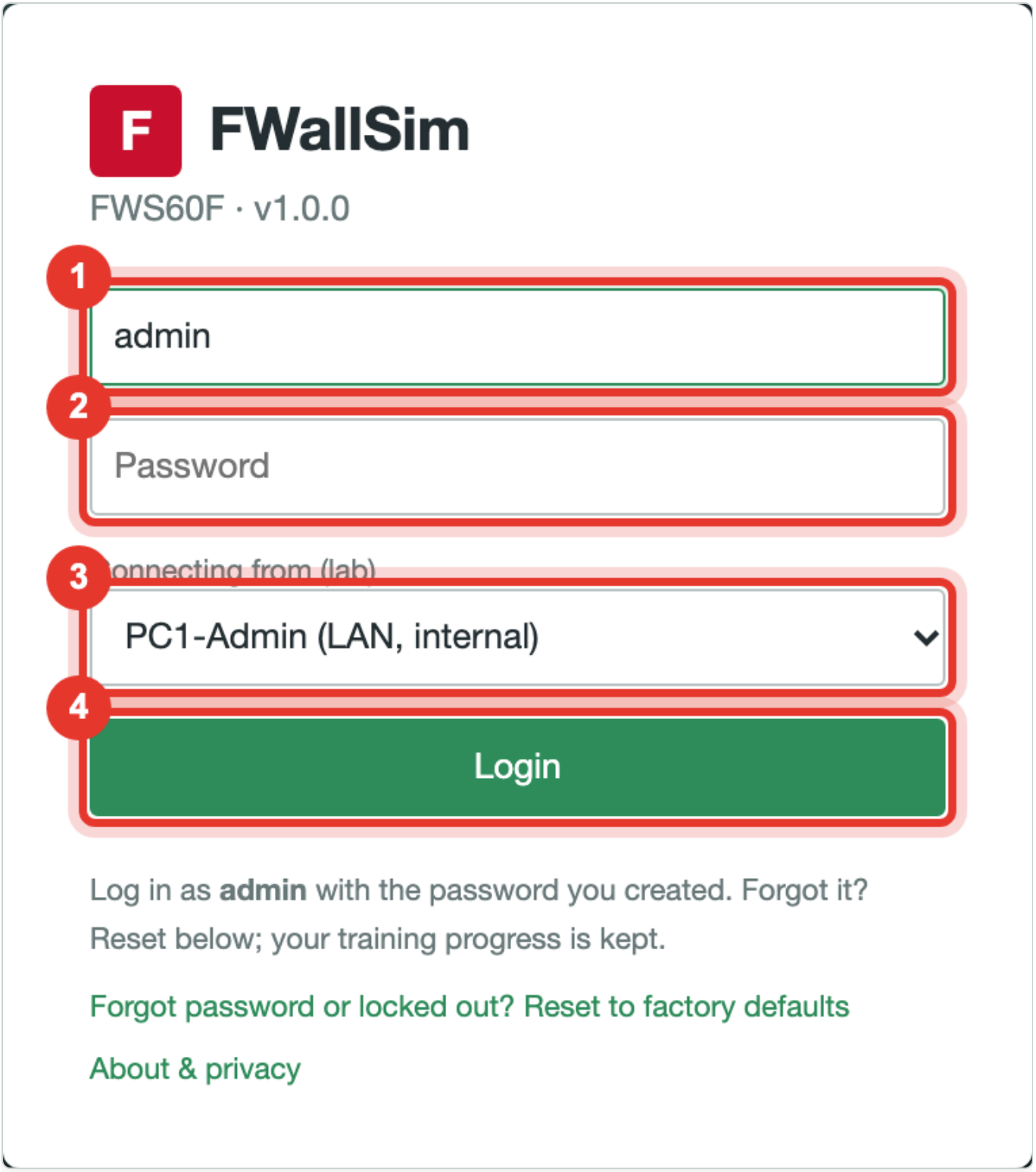
Create Password



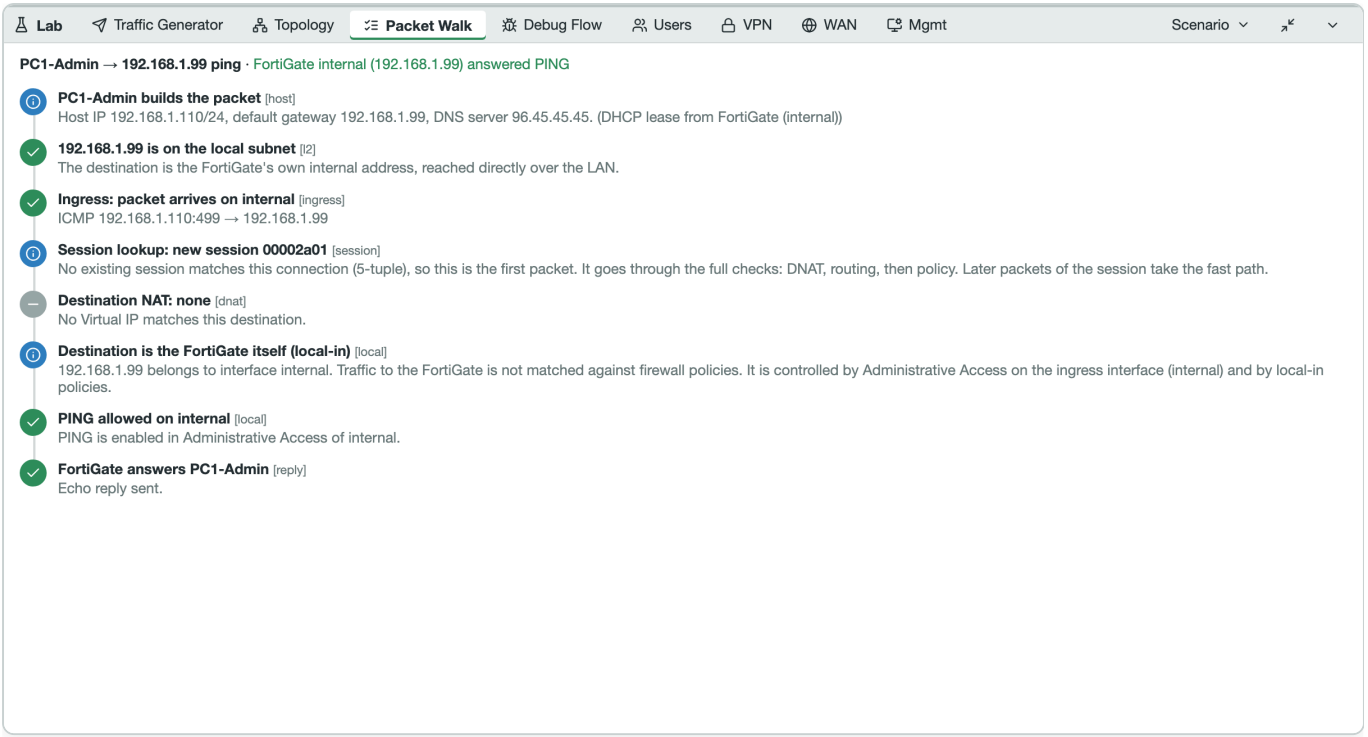
Dashboard



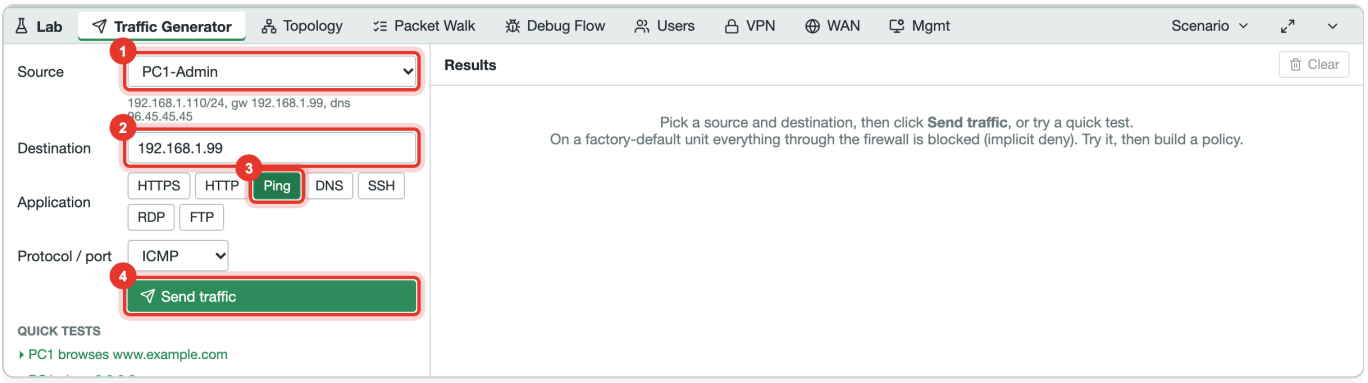
Header


FWallSim
FWS60F · v1.0.0
1
admin
2
Password
3
onnecting from (lab)
PC1-Admin (LAN, internal)
4
Login
Log in as admin with the password you created. Forgot it?
Reset below; your training progress is kept.
Forgot password or locked out? Reset to factory defaults
About & privacy

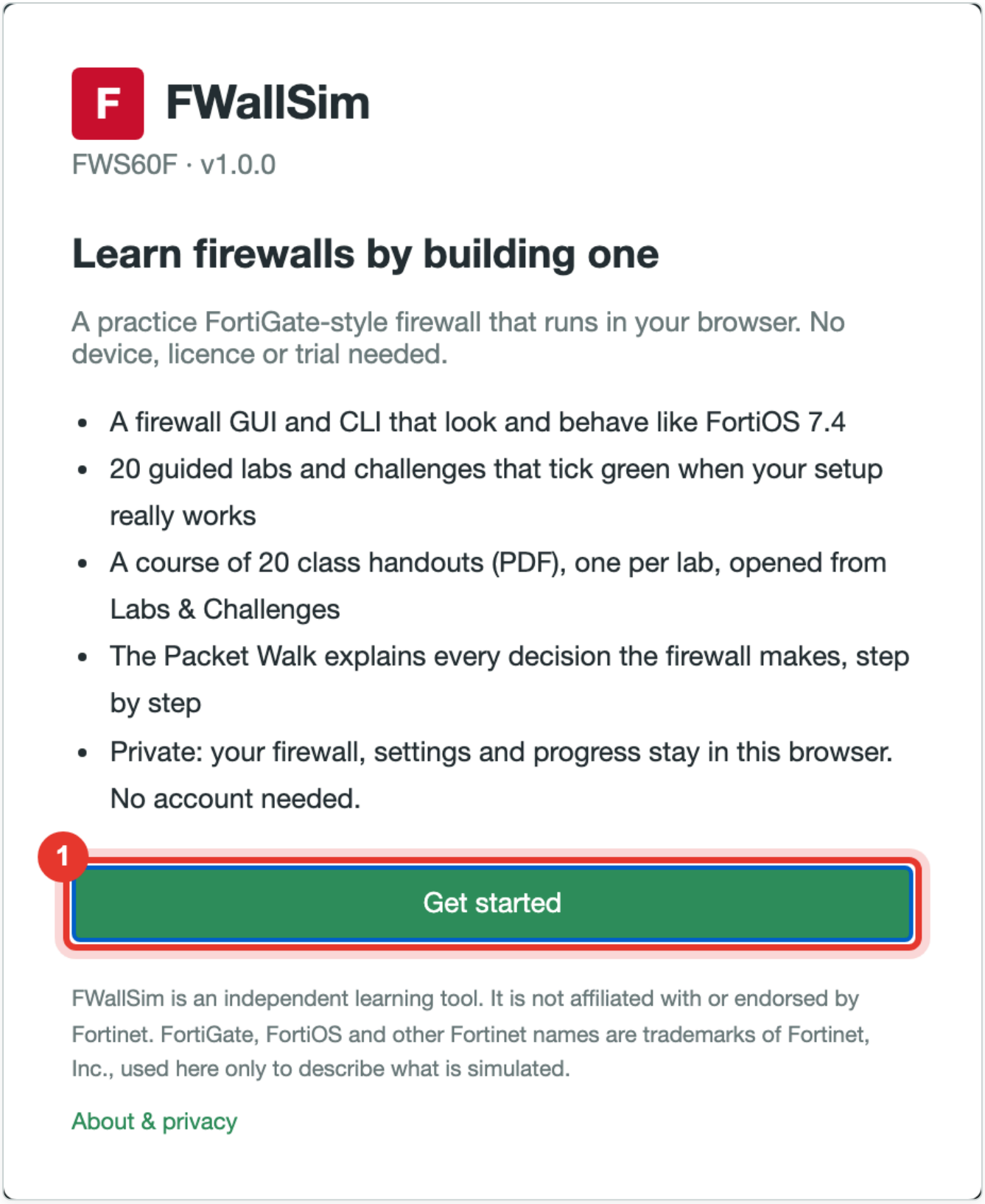
Login



Ping Walk



Traffic Gen


F
FWallSim
FWS60F · v1.0.0
Learn firewalls by building one
A practice FortiGate-style firewall that runs in your browser. No device, licence or trial needed.
A firewall GUI and CLI that look and behave like FortiOS 7.4
20 guided labs and challenges that tick green when your setup really works
A course of 20 class handouts (PDF), one per lab, opened from Labs & Challenges
The Packet Walk explains every decision the firewall makes, step by step
Private: your firewall, settings and progress stay in this browser. No account needed.
1
Get started
FWallSim is an independent learning tool. It is not affiliated with or endorsed by Fortinet. FortiGate, FortiOS and other Fortinet names are trademarks of Fortinet, Inc., used here only to describe what is simulated.
About & privacy

Welcome

CLI Console ? help · Tab complete · ↑↓ history · Ctrl+C stop · Ctrl+L clear

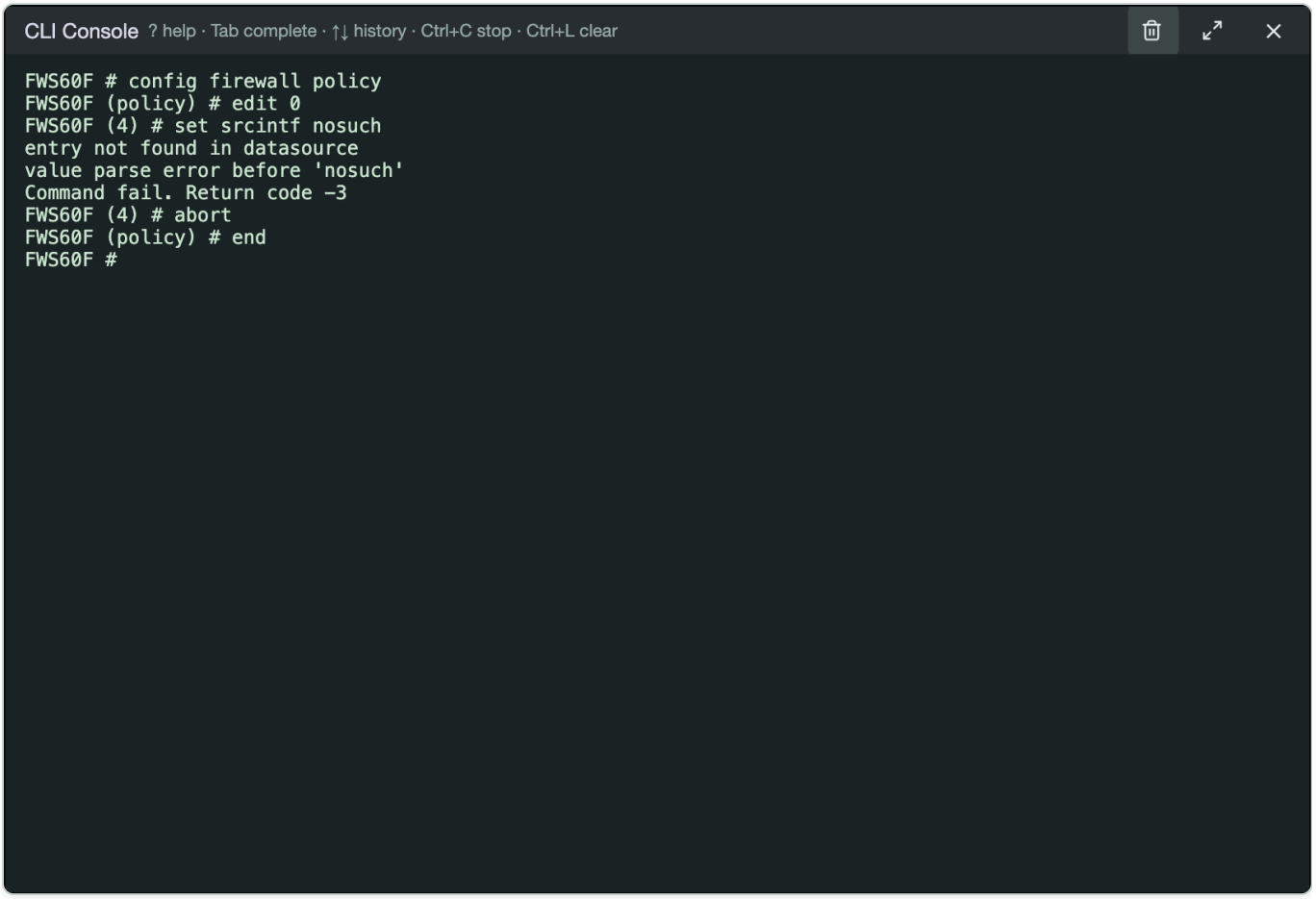
```
FWS60F # config system interface
FWS60F (interface) # edit internal
FWS60F (internal) # show
config system interface
    edit "internal"
        set ip 192.168.1.99 255.255.255.0
        set allowaccess ping https ssh fabric
        set type hard-switch
        set role lan
        set snmp-index 9
    next
end
FWS60F (internal) # end
FWS60F #
```

Config Mode

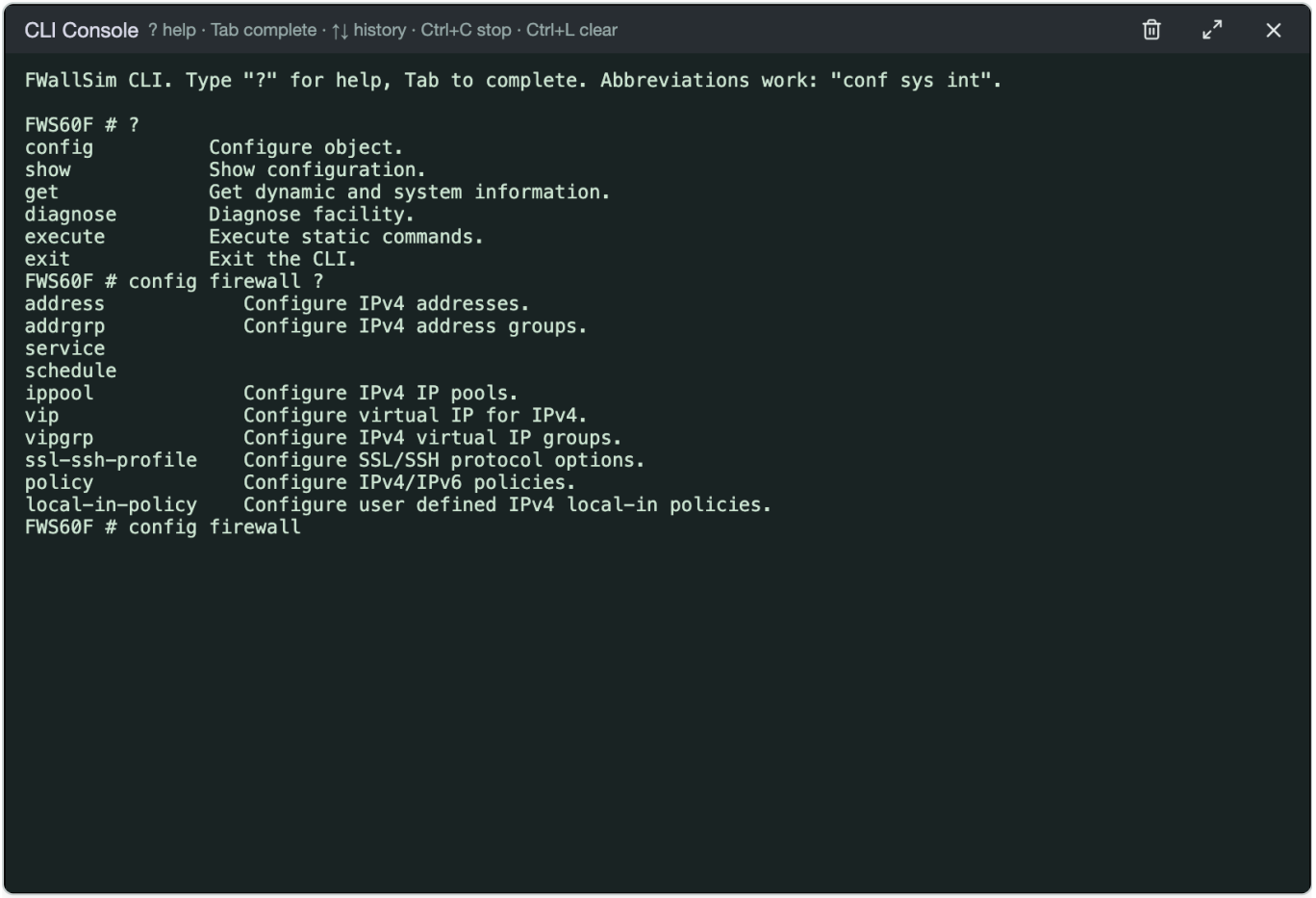
CLI Console ? help · Tab complete · ↑↓ history · Ctrl+C stop · Ctrl+L clear

```
FWS60F # diagnose debug flow filter addr 8.8.8.8
FWS60F # diagnose debug flow show function-name enable
FWS60F # diagnose debug flow trace start 10
(note: debug output is disabled. Run "diagnose debug enable" to see trace output.)
FWS60F # diagnose debug enable
id=65308 trace_id=1 func=print_pkt_detail line=5895 msg="vd-root:0 received a packet(proto=1, 192.168.1.110:4
4->8.8.8.8:2048) tun_id=0.0.0.0 from internal. type=8, code=0, id=44, seq=1."
id=65308 trace_id=1 func=init_ip_session_common line=6076 msg="allocate a new session-00002a01, tun_id=0.0.0.
0"
id=65308 trace_id=1 func=vf_ip_route_input_common line=2605 msg="find a route: flag=04000000 gw-203.0.113.1 v
ia wan1"
id=65308 trace_id=1 func=fw_forward_handler line=881 msg="Allowed by Policy-1: SNAT"
id=65308 trace_id=1 func=__ip_session_run_tuple line=3497 msg="SNAT 192.168.1.110->203.0.113.10:60299"
id=65308 trace_id=2 func=print_pkt_detail line=5895 msg="vd-root:0 received a packet(proto=1, 8.8.8.8:0->203.
0.113.10:2048) tun_id=0.0.0.0 from wan1. type=0, code=0, id=0, seq=1."
id=65308 trace_id=2 func=resolve_ip_tuple_fast line=5976 msg="Find an existing session, id-00002a01, reply di
rection"
id=65308 trace_id=2 func=__ip_session_run_tuple line=3511 msg="DNAT 203.0.113.10:60299->192.168.1.110:44"
id=65308 trace_id=2 func=vf_ip_route_input_common line=2605 msg="find a route: flag=00000000 gw-192.168.1.110
via internal"
FWS60F #
```

Debug Flow


CLI Console ? help · Tab complete · ↑↓ history · Ctrl+C stop · Ctrl+L clear
FWS60F # config firewall policy
FWS60F (policy) # edit 0
FWS60F (4) # set srcintf nosuch
entry not found in datasource
value parse error before 'nosuch'
Command fail. Return code -3
FWS60F (4) # abort
FWS60F (policy) # end
FWS60F #

Errors


CLI Console ? help · Tab complete · ↑↓ history · Ctrl+C stop · Ctrl+L clear
FWallSim CLI. Type "?" for help, Tab to complete. Abbreviations work: "conf sys int".
FWS60F # ?
config Configure object.
show Show configuration.
get Get dynamic and system information.
diagnose Diagnose facility.
execute Execute static commands.
exit Exit the CLI.
FWS60F # config firewall ?
address Configure IPv4 addresses.
addrgrp Configure IPv4 address groups.
service
schedule
ippool Configure IPv4 IP pools.
vip Configure virtual IP for IPv4.
vipgrp Configure IPv4 virtual IP groups.
ssl-ssh-profile Configure SSL/SSH protocol options.
policy Configure IPv4/IPv6 policies.
local-in-policy Configure user defined IPv4 local-in policies.
FWS60F # config firewall

Help

CLI Console ? help · Tab complete · ↑↓ history · Ctrl+C stop · Ctrl+L clear

```
                status: up
                speed: 1000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
        ==[internal2]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: up
                speed: 1000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
        ==[internal3]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: down
                speed: n/a
                FEC: none
                FEC_cap: none
        ==[internal4]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: down
                speed: n/a
                FEC: none
                FEC_cap: none
        ==[internal5]
                mode: static
                ip: 0.0.0.0 0.0.0.0
                ipv6: ::/0
                status: down
                speed: n/a
                FEC: none
                FEC_cap: none
FWS60F #
```

Show Get

CLI Console ? help · Tab complete · ↑↓ history · Ctrl+C stop · Ctrl+L clear

```
FWS60F # diagnose debug disable
FWS60F # diagnose sniffer packet any "host 8.8.8.8" 4
Using Original Sniffing Mode
interfaces=[any]
filters=[host 8.8.8.8]
(Send traffic from the Lab panel. Press Ctrl+C to stop.)
0.059085 internal in 192.168.1.110 -> 8.8.8.8: icmp: echo request
0.059281 wan1 out 203.0.113.10 -> 8.8.8.8: icmp: echo request
0.060386 wan1 in 8.8.8.8 -> 203.0.113.10: icmp: echo reply
0.060760 internal out 8.8.8.8 -> 192.168.1.110: icmp: echo reply
^C

4 packets received by filter
0 packets dropped by kernel

FWS60F #
```

Sniffer

Next: Class 2: Your first firewall policy: LAN to Internet